Management and Restoration of Invaded Ecosystems

FNR 6516 (3 credits) online
Spring 2026

Course Description
An overview of the ecological basis for plant invasions in terrestrial ecosystems, with primary emphasis on applications for restoration and management of invaded ecosystems. Methods and techniques for prediction, prevention, control, and restoration will be discussed, and plant invasions from Florida and around the U.S. will be used as case studies. This course focuses heavily on applying scientific theory and research to on-the-ground management.

Format and Audience
This course will follow an online asynchronous discussion format, with recorded lectures and relevant assigned readings from textbooks and primary literature. The course is graduate level and is designed for students with a strong interest and background in ecology and applied plant science and an interest in invasive species ecology and management. **The first half of the semester is designed to have less instructor interaction than the second half of the semester.**

Pre-Requisites
No formal pre-requisites, but coursework in biology, ecology, or other relevant plant science courses is strongly recommended.

Instructors

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TA's

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Office Hours
Given that this course is remotely taught the best way to set up a meeting is by emailing mandreu@ufl.edu and we can arrange to set up a Zoom/TEAMS call.

Group Meetings with the Instructors
Throughout the semester we will schedule meetings with the instructors and all students enrolled in the course. Participation in these meetings is optional. Times will be listed in CANVAS.

Learning Outcomes
At the end of this course, each student will be able to:

- Critically assess scientific literature and implications of results for practical management.
- Evaluate invasive species documented on a site and identify potential impacts.
- Describe the desired future ecosystem conditions incorporating available resources.
- Identify infestations that impact the current condition.
- Incorporate knowledge of invasion theory and mechanisms to design and prioritize treatments in an annual and/or multi-year adaptive management plan.
- Propose a monitoring scheme to support an adaptive management approach.

Required Text
1. Invasion Ecology 2nd ed. JL Lockwood, MF Hoopes and MP Marchetti. 2013. Blackwell Publishing,303
p. 978-1444333657
A virtual link to the textbook Invasion Ecology (2nd edition) is available if you have a UFID.

Class Format
The course will consist of one-week modules focused on specific topics related to invasion ecology, management, and restoration. The format will consist primarily of readings and discussion threads. To accommodate students with full-time employment, modules will follow a Friday-Sunday (10 days, other than week 1) schedule to allow time for adequate discussion over the weekend period as needed. For each module in the first half of the course, students will be assigned several readings, including chapter(s) from one of the required texts, relevant peer-reviewed journal articles, or other materials. A short (approximately 20 minute) summary lecture to review core lessons from the general topic will be provided by the instructors. The lecture will be posted each Friday. Throughout the semester, some additional guest lectures and video podcasts will be provided as a supplement. **The first half of the semester is designed to have less direct instructor interaction than the second half of the semester.** However, do not hesitate to reach out to the instructors with any questions or for any needed help with the course materials or structure.

The second half of the course will take an interrupted case study format, with several relevant, peer-reviewed journal articles to introduce the general topic, plus a short description of that module's section of the case study and relevant questions with a discussion thread.

A discussion thread will also be posted on Friday. Nine discussions will be led by the instructors (focusing on the module topic) and five discussions will be led by a group of students (focusing on a journal article). These additional readings will build on topics introduced in the lectures and/or present a case study of relevant invasive plant ecology and management. All students are expected to read these articles and participate in the additional discussion. Comments/responses from the students can be posted until Saturday evening. **Students are encouraged to post early and often if they want to be fully engaged in the course.**

NOTE: *Discussion questions are intended to stimulate conversation and debate and encourage you to explore more deeply the topics covered in the week's readings. In many cases, there will not be a clear "right" or "wrong" answer. In some cases, the questions will be contextual (e.g., "Describe an example of a species that exhibits invasive traits"), other questions will be more conceptual, and some questions may ask to merely express an opinion. Towards the end of the semester the discussion threads will be used to practice developing adaptive management recommendations for an invaded ecosystem.*

Late policy for assignments and attendance: "Attendance" for this course will be based on participation in the discussion forum. Written assignments and projects are due electronically by midnight (Eastern time) on the due date and **unexcused late work will lose 10% of the grade for each day they are late** (weekends count too). In cases of extended illness or emergencies, arrangements to turn in late exams or other written assignments must be made with the instructor prior to the due date. Requirements for class attendance and make-up exams, assignments and other work are consistent with university policies that can be found at: https://gradcatalog.ufl.edu/graduate/regulations/.

Assignments and Evaluation of Student Learning

Discussion thread participation

Students will be expected to contribute **two** unique comments and/or responses to other students (typically several sentences to about a paragraph in length) which demonstrate thought and/or research into the topic area. A citation relevant to at least one of your comments is required. Note that you are welcome to post and respond more than the minimum.

Rubric:
0 Points: No response
.5 Points: only one comment that demonstrates thought and/or research.
.8 Points: 2 comments, no citation or 1 comment with citation.
1.0 Points: 2 comments that demonstrate thought and/or research including relevant citation.

Student-led Discussion thread

For the student-led discussions, the discussion leader(s) will be expected to read the article (and supporting literature, as necessary) and lead a discussion on the most important topics covered in it. This will involve providing a brief 1-2 paragraph summary, posing at least 3 questions for the other students, and *facilitating* a productive online dialogue between students. The discussion leader(s) should follow the directions in the relevant Canvas assignment, which are typically to email the article summary and discussion questions to a TA by the Thursday evening before the discussion opens.

Rubric for group leaders: While it is important that all group members contribute to the 1) development of questions, 2) writing of the article summary, 3) moderation of the discussion, and 4) providing a post-discussion summary with key points; all members do not have to do all four things equally but should be significant contributors to at least 2 of the four areas.

Individual score: Individual contribution to each of the three areas: 16 points.

Group score: Group will collectively be scored for the overall week: 12 pts

Summary of papers was insightful, succinct yet complete: Y/N
The discussion prompt questions stimulated thoughtful discussion: Y/N
Moderators actively encouraged cogent responses: Y/N
Post-discussion summary ties together salient points to that module's themes: Y/N

Management Plan Project

You will develop and present an **actionable** management plan for restoring and managing a particular property with non-native species invasions. You are encouraged to choose a property that is local to you and/or that you are currently working on; or you can work with the instructors to find a suitable scenario relative to your locale. Your management plan should provide an overview of the non-native species of concern. It should include mechanisms for dispersal into your site and ecosystem impacts, a feasible annual work plan for control of the current invasion, restoration of ecological characteristics (e.g., species composition, structure, soils/hydrology, or other ecological processes) following control, and monitoring and prevention of new invasions. Prioritization of actions should also be discussed.

Two homework assignments during the second half of the semester will build up to the final paper (invasive species management plan), allowing for instructor input and increased application of knowledge by the student.

As part of your grade, you will also be asked to peer review plans presented by your fellow students. More detailed instructions on this assignment and directions for uploading your materials will be provided in the Assignments tab of Canvas.

The grading breakdown will be as follows:

- 14 points Participation in weekly discussion sessions (1 point each *x* 14 discussions)
- 28 points Presentation of one weekly article and moderation of discussion (Group)
- 10 points Quizzes (2 points x 5 quizzes)
- 20 points Homework assignments leading to management plan (2 x 10 points each)
- 20 points Management plan project.
- 8 points Peer review of student management plan (2 x 4 points each)
- Total: 100 points

Grading Scale

Letter grades will be assigned as follows: A (93-100), A^- (90-92.99), B^+ (86-89.99), B (83-85.99), B^- (80- 82.99), C^+(76-79.99), C (73-75.99), C^-(70-72.99), D^+(66-69.99), D (63-65.99), D- (60-62.99), E (<60)

Schedule of Class Topics and Readings

Introduction (Why Invasive Species Are a Problem)

Module 1: Introduction

Required Text: Lockwood et al, Chapter 1, An Introduction to Invasion Ecology
Article: van Kleunen, Mark, Oliver Bossdorf, and Wayne Dawson. "The ecology and evolution of alien plants." *Annual Review of Ecology, Evolution, and Systematics* 0 (2018).
Article: Ricciardi, Anthony, and Rachael Ryan. "The exponential growth of invasive species denialism." *Biological Invasions* 20, no. 3 (2018): 549-553.
Article: Sagoff, Mark. "Invasive species denialism: a reply to Ricciardi and Ryan." *Biological Invasions* (2018): 1-7.

Module 2: Critically Evaluating Invasive Species Literature

Article: Frazier, Jesse E., Ajay Sharma, Daniel J. Johnson, Michael G. Andreu, and Kimberly K. Bohn. "Group selection silviculture for converting pine plantations to uneven-aged stands." *Forest Ecology and Management* (2020): 118729.

Module 3: Impacts to Individuals, Species and Communities

Required Text: Lockwood et al, Chapter 9 through page 233, Ecological Impacts of Invasive Species
Article: Liebhold, Andrew M., Eckehard G. Brockerhoff, Susan Kalisz, Martin A. Nuñez, David A.Wardle, and Michael J. Wingfield. "Biological invasions in forest ecosystems." Biological invasions 19, no. 11 (2017): 3437-3458.
A*rticle*: Kamaru, Douglas N., Todd M. Palmer, Corinna Riginos, Adam T. Ford, Jayne Belnap, Robert M. Chira, John M. Githaiga et al. "Disruption of an ant-plant mutualism shapes interactions between lions and their primary prey." *Science* 383, no. 6681 (2024): 433-438.

Module 4: Impacts to Ecological Processes and Economics

Required Text: Lockwood et al, finish Chapter 9, Ecological Impacts of Invasive Species
Article: Fantle-Lepczyk, Jean E., Phillip J. Haubrock, Andrew M. Kramer, Ross N. Cuthbert, Anna J. Turbelin, Robert Crystal-Ornelas, Christophe Diagne, and Franck Courchamp. "Economic costs of biological invasions in the United States." Science of the Total Environment 806 (2022): 151318.
Student-led *article*: Essl, Franz, Philip E. Hulme, Jonathan M. Jeschke, Reuben Keller, Petr Pyšek, David M. Richardson, Wolf-Christian Saul et al. "Scientific and normative foundations for the valuation of alien-species impacts: thirteen core principles." AIBS Bulletin 67, no. 2 (2016): 166-178.

Invasion Theory (How They Become and Cause These Problems)

Module 5: Dispersion and the Invasion Process
Required text Lockwood et al. Chapters 2, Transport Vectors and Pathways; and Chapter 4, Propagules.
Article: Harvey, Rebecca G., and Frank J. Mazzotti. "The invasion curve: A tool for understanding invasive species management in south Florida." IFAS Publication Number WEC347. Gainesville, FL: University of Florida. edis. ifas. ufl. edu/uw392(2014).
Student-led *article:* Daly, Ella Z., Olivier Chabrerie, Francois Massol, Benoit Facon, Manon CM Hess, Aurélie Tasiemski, Frédéric Grandjean et al. "A synthesis of biological invasion hypotheses associated with the introduction–naturalisation–invasion continuum." Oikos 2023, no. 5 (2023): e09645.
Optional text: Lockwood et al. Chapter 8, Ecological Processes and the Spread of Non-native Species

Module 6: Disturbances and How They Impact Invasions
Required Text: Lockwood et al, Chapter 5, Disturbance; and Chapter 6, Establishment Success: The Influence of Biotic Interactions

Article: Xiao, Sa, Ragan M. Callaway, Ryan Graebner, Jose L. Hierro, and Daniel Montesinos. "Modeling the relative importance of ecological factors in exotic invasion: The origin of competitors matters, but disturbance in the non-native range tips the balance." Ecological modelling 335 (2016): 39-47.
Student-led *article*: Pearson, Dean E., Yvette K. Ortega, Diego Villarreal, Ylva Lekberg, Marina C. Cock, Özkan Eren, and José L. Hierro. "The fluctuating resource hypothesis explains invasibility, but not exotic advantage following disturbance." Ecology 99, no. 6 (2018): 1296-1305.

Management Planning (How to Limit or Remove These Problems)

Module 7: Management Planning- Assessment (What Do You Have?)
Article: Gordon, Doria R., S. Luke Flory, Deah Lieurance, Philip E. Hulme, Chris Buddenhagen, Barney Caton, Paul D. Champion et al. "Weed risk assessments are an effective component of invasion risk management." *Invasive Plant Science and Management* 9, no. 1 (2016): 81-83.
Article: Lieurance, D. "Protocols for testing the invasiveness of plants in Florida." In *Proceedings of the 2015 Annual Meeting of the International Plant Propagators' Society 1140*, pp. 279-284. 2015.
Optional text: Lockwood et al. Chapter 12, Predicting and Preventing Invasion

Module 8: Management Planning- Assessment (Biology and Control)
A. *Required Text*: Chapter 13, Lockwood et al. Ecological Processes and the Spread of Non-native Species

Module 9: Discussion, HOMEWORK ASSIGNMENT
Student-led *article*: Pecl, Gretta T., Miguel B. Araújo, Johann D. Bell, Julia Blanchard, Timothy C. Bonebrake, I-Ching Chen, Timothy D. Clark et al. "Biodiversity redistribution under climate change: Impacts on ecosystems and human well-being." *Science* 355, no. 6332 (2017).
B. *Optional article*: Beaury, Evelyn M., Emily J. Fusco, Michelle R. Jackson, Brittany B. Laginhas, Toni Lyn Morelli, Jenica M. Allen, Valerie J. Pasquarella, and Bethany A. Bradley. "Incorporating climate change into invasive species management: insights from managers." *Biological Invasions* 22, no. 2 (2020): 233-252.

Module 10: Management Planning- Desired Future Conditions (What Do You Want?)
A. A*rticle:* Messier, Christian, Klaus Puettmann, Robin Chazdon, K. P. Andersson, Virginie A. Angers, L. Brotons, E. Filotas, Rebecca Tittler, Lael Parrott, and Simon A. Levin. "From management to stewardship: viewing forests as complex adaptive systems in an uncertain world." *Conservation Letters* 8, no. 5 (2015): 368-377.
B. *Optional text*: Lockwood et al. Chapter 14, Global Climate Change and Invasive Species

Module 11: Peer Review, HOMEWORK ASSIGNMENT
Online readings posted on Canvas
Student-led *article*: Wallingford, Piper D., Toni Lyn Morelli, Jenica M. Allen, Evelyn M. Beaury, Dana

M. Blumenthal, Bethany A. Bradley, Jeffrey S. Dukes et al. "Adjusting the lens of invasion biology to focus on the impacts of climate-driven range shifts." *Nature Climate Change* (2020): 1-8.

Module 12: Management Planning- Building a Treatment Plan (How Do You Get There?)
Article: Stone, Deborah, and Michael Andreu. "Direct Application of Invasive Species Prioritization: The Spatial Invasive Infestation and Priority Analysis Model." *Ecological Restoration* 35, no. 3 (2017): 255-265.

Module 13: Management Planning- Finalizing and Implementing Your Management Plan (Adaptive Management)
Article: Prior, Kirsten M., Damian C. Adams, Kier D. Klepzig, and Jiri Hulcr. "When does invasive species removal lead to ecological recovery? Implications for management success." *Biological invasions* 20, no. 2 (2018): 267-283.
Optional article: Stone, Deborah, and Michael Andreu. "Fire and Invasive Plant Interactions" University of Florida IFAS Extension FOR 386 (2022) 13 pp. https://doi.org/10.32473/edis-FR457-2022

Module 14: FINAL PROJECT (See CANVAS for due dates)

Academic Policies and Resources

Academic policies for this course are consistent with university policies. See https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/

Campus Health and Wellness Resources

Visit https://one.uf.edu/whole-gator/topics for resources that are designed to help you thrive physically, mentally, and emotionally at UF.
Please contact UMatterWeCare for additional and immediate support.

Software Use

All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Privacy and Accessibility Policies

[required for online courses, list all technology used]

- Instructure (Canvas)
  - Instructure Privacy Policy
  - Instructure Accessibility
- Zoom
  - Zoom Privacy Policy
  - Zoom Accessibility